

Joel E. Castro Hernandez

Los Angeles, CA | joelcast@usc.edu | (619) 610-8132 | www.linkedin.com/in/joel-castro- | <https://joel-ca.github.io/portfolio/>

EDUCATION

University of Southern California, Viterbi School of Engineering

M.S. Computer Science

August 2026 - May 2028

Los Angeles, CA

UC Berkeley, College of Letters & Sciences - GPA: 3.5

B.A. Computer Science

July 2022 - May 2026

Berkeley, CA

Scholarships: Jack Kent Cooke, SEED Scholar, CODE2040 Fellow, eBay Pathways Fellow, Cal Alumni Association Leadership Award

Relevant Coursework: Discrete Math & Probability, Computer Architecture, 3D Modeling & Animation, Signals & Systems, Robotic Manipulation and Interaction, Computer Graphics, Machine Learning, Efficient Algorithms & Intractable Problems, Computer Vision

SKILLS

Languages: Python, C/C++, Java, Bash, SQL, JavaScript/HTML/CSS

Software & Tools: ROS, Unreal Engine, Git, Arduino, Logisim

Design: Autodesk (Maya, Fusion, Inventor), Adobe (Ps, Pr, Ae, Substance), Nomad Sculpt

Spoken Languages: English & Spanish (Native/Bilingual)

EXPERIENCE

REAL Labs, Stanford

Stanford SURF REU, Summer Research Intern

March 2025 – August 2025

Stanford, CA

- Co-authored (in draft) with Professor Shuran Song and Stanford graduate students to train and deploy single- and multi-task Diffusion Policies toward establishing a goal-image-conditioned baseline to support future comparison with novel policy architectures.
- Designed a novel robotic handwriting task, including a 3D-printed pen gripper, for handheld data collection (collecting 450+ demonstrations), transferred to ARX5 robot for rollout.
- Built an evaluation pipeline leveraging Grounded-SAM2 and OpenCV for benchmarking of handwriting performance for publication.

Cardiac Vision Laboratory, UC San Francisco

Research Assistant

September 2024 – January 2025

San Francisco, CA

- Advanced 3D image processing methods to optimize visualization techniques for cardiac optical and ultrasound imaging data.
- Supported Professor Jan Christoph and Post-Doc Jan Lebert by developing a multi-view camera calibration and visualization tool, automating the calibration process, eliminating the need for repeated manual execution, and minimizing downtime by 90%.

Carnegie Mellon University Software and Societal Systems Department

REUSE SWE, Summer Intern

May 2024 – August 2024

Pittsburgh, PA

- [First-authored](#) a type theory for enhancing diagramming tools and explored a type system for diagrams.
- Categorically coded 150+ open-source visualization tools' source code (e.g., AMD GPUOpen, Torchview) for their data structure decomposition properties, yielding statistically significant evidence supporting our proposed diagram type theory.

Center for Computational Biology, UC Berkeley, Stellar Labs

Computational Biologist, Undergraduate Research Intern

August 2022 – May 2024

Berkeley, CA

- Exploring SHAP analysis on an in-lab convolutional neural network to increase its interpretability by determining which features it weighs greatest when predicting the presence of activation domains in proteins, leading to a [research publication](#) (in pre-print).
- Streamlined all-atom Monte Carlo simulation pipeline of disordered proteins on a computer cluster with Bash scripting, leading to a time-saving increase of over 85%, and wrote corresponding documentation used in training 2 peers.

PROJECTS

Minecraft Fabric Mod – CS184 Final Project Honorable Mention | [Website](#)

January 2025 – May 2025

- Implemented procedural sphere generation, texture mapping, and quaternion-based rotation mechanics for physically accurate interactions.
- Contributed to shader enhancements, video production, and final deliverables, helping the project earn Honorable Mention in CS184.

Sockrates: Clothing color sorting/folding on 7-degree-of-freedom industrial robot arm | [Website](#)

August 2024 – December 2024

- Led the development of Python/OpenCV categorization scripts, resulting in 100% classification accuracy.
- Built ROS Publisher/Subscriber System: integrated CV and precise robotic actuation implementing forward and inverse kinematics.

LEADERSHIP & VOLUNTEERING EXPERIENCE

UC Berkeley Undergraduate Graphics Group (UCBUGG)

Facilitator

January 2025 – May 2026

Berkeley, CA

- Presented weekly lectures on Autodesk Maya, Pixar's Renderman, and Adobe After Effects skills for a 3-unit course.
- Grade labs, host office hours, and mentor student groups through the end-to-end production of animated shorts.

(Berkeley) Anova

Onsite tutor; Publicity/Curriculum Committee Member

January 2023 – May 2026

Berkeley, CA

- Teach coding concepts in Python and Scratch to high school students weekly, enhancing their understanding of computer science.
- Developed and implemented Arduino curricula to improve tech accessibility for students in under-resourced communities.